

CRR retrodiction battery — outcomes

Independent run per the retrodiction prompt. One script (crr_retrodictions.py), no free parameters, every number printed by the script. Systems chosen per class by the runner; grades per the five-verdict scheme. Cross-listed rows (a system that answers in two classes) are counted once.

Tally (unique systems: 42)

SHARP	CONSIST	DESCR	FAILS	TENSION	OPEN
16	12	9	2	1	2

By class

class	SHARP	CONSIST	DESCR	FAILS	TENSION	OPEN
(a) Two-state / occupancy families	7	0	1	0	0	0
(b) Quantum states and dynamics	3	1	2	0	0	0
(c) Thermal ensembles / finite-time thermodynamics	3	1	1	0	0	1
(d) Parametric estimation and filtering	1	4	0	1	0	0
(e) Oscillators and limit cycles	0	1	2	1	0	1
(f) Point processes / natural time	1	4	0	0	0	0
(g) Gravitational / astrophysical / cosmological	0	0	2	0	0	0
(h) Bifurcations and critical phenomena	1	1	1	0	1	0

Reading

Where the framework is sharp: on families whose Fisher geometry is the physics' own. Every two-state occupancy law tested (half-life, logistic inflection, K_M /Hill, pKa, Fermi level, Langmuir, laser threshold) is the single statement 'the cut is at occupation one-half', forced by the Bernoulli family's half-turn with nothing supplied from outside. In quantum dynamics the cut rule reproduces the Mandelstam–Tamm speed limit and gives the actual cut time for spin-j and neutrino carriers. In thermal physics A1's admissibility condition ($\text{Var}(E) > 0$) rejects the Schwarzschild and de Sitter thermal families outright, and D2 coincides with thermodynamic length so that S is the excess length of a sub-optimal protocol. D8 turns critical slowing down into divergent memory depth.

Where it is not: oscillators. Class (e) produced no SHARP row. The phase arc of a rotor is π per half-turn by definition, so content must be read from an observable whose metric the framework has to choose; the equal-half-turn-arcs claim fails on every asymmetric cycle (FHN 0.945, biased Duffing 0.934) and holds on van der Pol only by symmetry; the cut placement on a relaxation oscillator is an assignment, not a derivation. Classes (d) and (f) are almost entirely CONSIST: the framework's forms coincide with Kalman, Cramér–Rao, Poisson and Omori but fix no constant.

Failures: the equanimity gain $K(1)=1/\phi$ is a random-walk result and drifts (0.60 to 0.50) as mean reversion strengthens; equal arms is a symmetry, not a theorem. Open: the cost of a cut (Landauer) and the length L of a rotor's carrier.

Tension and proposed rewording: 'criticality = maximal openness, ρ to 0' is true at a Hopf point and false at a second-order thermal transition, where the Fisher metric diverges with the correlation length and ρ to infinity. Minimal fix: state criticality through D8 (memory depth $d = 1/K(v)$ diverges as the relaxation rate vanishes), which holds for both, and confine the ρ -to-0 statement to the Hopf case.

All rows

cls	system	clause	verdict	printed numbers	note
a	First-order decay (radioactive, unimolecular)	A3 antipode on Bernoulli family	SHARP	Fisher half-turn at $p=[1/2]$; t at $p=1/2$ is $\ln 2/k$	the cut is the half-life; forced, no constant supplied
a	Logistic growth	A3 antipode	SHARP	antipode $p=1/2$ = inflection (max dN/dt at $N=K/2$)	forced by the same family
a	Michaelis–Menten and Hill binding	A3 antipode	SHARP	$\theta=1/2$ at $S=[K_M]$ for every Hill n	$K_M / K_d / K_{half}$ are the antipode; cooperativity does not move it
a	Acid–base titration (Henderson–Hasselbalch)	A3 antipode	SHARP	ionised fraction $1/2$ at $pH = pK_a$	half-equivalence point is the cut
a	Fermi–Dirac occupation	A3 antipode	SHARP	$f(E=\mu) = 1/2$	the chemical potential is the antipode of the occupation family
a	Thermal two-level system / laser threshold	A3 antipode	SHARP	equilibrium $p_{upper} \rightarrow 1/2$ only as $T \rightarrow \infty$ ($\lim = 1/2$); crosses $1/2$ only under pumping	no thermal cut at finite T ; population inversion = crossing the antipode = laser threshold
a	Langmuir adsorption	A3 antipode	SHARP	$\theta = 1/2$ at $P = 1/K_{ads}$	half-coverage pressure is the cut (same family as Michaelis–Menten)
a	Ising mean-field magnetisation	A3 antipode	DESCR	$m=0 \Leftrightarrow$ up-fraction $1/2$ at $T \geq T_c$; below T_c the family sits off the antipode	the disordered phase is the antipode of the spin-occupation family; true, but definitional
b	Qubit precession (Rabi/Larmor)	A3 cut time = half-turn / Fisher speed	SHARP	$t_{orth}=2.4151$, $\pi\hbar/(2dE)=2.4166$	reproduces the Mandelstam–Tamm quantum speed limit, saturated
b	Spin- j coherent state under rotation	A3	SHARP	antipode at rotation angle π for all j : $t=\pi/\omega$; MT bound $\pi/(\omega \sqrt{2j})$ saturated only at $j=1/2$	framework gives the actual cut time; MT gives a bound; both agree at $j=1/2$
b	Two-flavour neutrino oscillation	A3 antipode reachability	SHARP	max conversion $\sin^2(2\theta)=[0.0.262 \ 0.756 \ 1.]$	flavour state reaches its Bures antipode iff mixing is maximal
b	Coherent-state displacement	A3 no antipode	DESCR	$ \langle 0 \alpha\rangle ^2 = e^{- \alpha ^2} > 0$ for all α	never orthogonal, never cuts
b	Quantum Zeno effect	A3 reset before the antipode	CONSIST	survival after $n=1000$ resets in one half-turn: 0.9975	cuts arriving before the antipode never complete an occasion; the framework does not derive the $(t/n)^2$ law
b	Decoherence (T_2) / mixed states	A3 no antipode for mixed states	DESCR	Bures antipode exists only for pure states	occasions stop completing at the decoherence time
c	Finite-time thermodynamics (thermodynamic length, Crooks 2007)	D2 = thermodynamic length; S = excess length	SHARP	Cauchy–Schwarz: $\langle v^2 \rangle = 1.084 \geq \langle v \rangle^2 = 0.724$; equality iff constant Fisher speed	minimum dissipation path is the geodesic at constant Fisher speed; S^2 bounds excess dissipation
c	Ideal-gas thermometer	A1' resolution	CONSIST	resolvable temperature steps = $\log((T_2/T_1)^{**}(\sqrt{6})\sqrt{N}/2))$	$\sqrt{N}$ resolution; standard Fisher metric of the canonical ensemble
c	Ideal gas (Ruppeiner)	A1 carrier	DESCR	Ruppeiner curvature $R = 0$	flat carrier, no interactions; consistent, uninformative
c	Schwarzschild thermal family	A1 admissibility ($\text{Var}(E) = C_V T^2 > 0$)	SHARP	$g_{bb} = -1/(\hbar^2 \pi) < 0$ for all M	Hawking's thermal family is not a carrier; only the unitary/microstate description is admissible
c	de Sitter horizon (Gibbons–Hawking)	A1 admissibility	SHARP	$C = T dS/dT = -2\pi\hbar/H^2 < 0$	static-patch thermal description likewise rejected as a carrier
c	Landauer erasure	cost of a cut	OPEN	not derived	whether a cut costs $kT \ln 2$ per resolvable step is not fixed by the axioms
d	Random-walk Kalman filter	P4 + A9	CONSIST	$K(1) = -1/2 + \sqrt{5}/2$	golden-ratio gain at $q=r$ is textbook; framework supplies the reading only
d	Mean-reverting (Ornstein–Uhlenbeck) tracking	A9 as a gain law	FAILS	optimal K by ϕ : {0.99: 0.6000000000000001, 0.6: 0.55, 0.2: 0.5} vs $1/\phi=0.618$	equanimity gain holds only in the random-walk limit; off it the state model sets the gain
d	Cramér–Rao bound	A1' unit	CONSIST	$\sigma = 1/\sqrt{I}$	identity
d	$\sqrt{2KL}$ as Fisher–Rao length	D2 for parametric models	CONSIST	fixed-variance Gaussian: $\sqrt{2\mu^2/2} = \text{Abs}(\mu)$	second-order identity, standard

cls	system	clause	verdict	printed numbers	note
d	Critical slowing down -> memory depth	D8	SHARP	$k=[1.0.30.10.030.01] \rightarrow d=[1.62.43.76.310.5]$	depth diverges as the relaxation rate vanishes; long memory near criticality, derived not fitted
d	Poisson rate estimation	A1 Fisher metric $g=1/\lambda$	CONSIST	$L(1 \rightarrow 4) = 2 = 2(\sqrt{4} - \sqrt{1})$	distinguishable rates scale as $\sqrt{N}$ (Wootters); shot noise
e	van der Pol vs FitzHugh–Nagumo vs biased Duffing — equal half-turn arcs	symmetry check on A3	FAILS	arm ratio $vdP=1.001$, $FHN=0.945$, $Duffing(biased)=0.934$	equal arms hold only for $x \rightarrow -x$ symmetric cycles; not a consequence of the cut
e	Isochronous pendulum vs integrate-and-fire	D2 regularity of arc vs clock	DESCR	pendulum $CV(arc)=0.28 > CV(clock)=0$; IAF $CV(arc)=0.00 < CV(clock)=0.23$	instrument recovers the known dichotomy; nothing derived
e	Relaxation oscillator (sawtooth) — cut placement	A3 on a rotor	OPEN	one event per turn; the phase antipode falls mid-ramp where nothing happens	on a rotor A3 does not fix L; whether an inter-event interval is a half-turn or a turn is an assignment (O3)
e	Hopf / Stuart–Landau normal form	criticality clause, $\rho \sim$ amplitude	CONSIST	amplitude $\sim \sqrt{\mu} \rightarrow \rho \sim \sqrt{\mu}$	square-root law inherited from the normal form
e	Kepler orbit	symmetry check on A3	DESCR	perihelion/aphelion arms equal by time-reversal	symmetry, not framework
f	Aftershock decay (Omori)	P5 natural-time retention	CONSIST	$e^{-A/\rho}$ is a power law in clock time iff $p=1$	consistency relation; p not derived
f	Black-hole evaporation in emitted quanta	A1' + A3 (half the steps)	SHARP	quanta fraction == entropy fraction: True; cut at $t/t_{ev}=0.6464$	the natural-time antipode is the Page point
f	Radioactive counting / photon shot noise	A1' natural time	CONSIST	rate resolution after N counts $\sim \sqrt{N}$ steps	identity with the Poisson metric
f	Black-hole scrambling (Sekino–Susskind)	A1' in quanta	CONSIST	$t^* = (\beta/2\pi) \ln S \sim \ln S$ steps	form only; prefactor not derived
f	Dieterich rate-and-state aftershocks	P5	CONSIST	rate-and-state gives $p=1$ independently; matches the natural-time exponential	same relation as Omori, from a mechanism the framework lacks
g	Schwarzschild (again, as carrier)	A1	SHARP	see class c: thermal family inadmissible	forces the unitary side
g	Page curve	A3 in natural time	SHARP	see class f	cut = Page point
g	Solar dynamo (Hale/Schwabe)	A3 two cuts per turn; A6 depth-one seed	DESCR	polarity reversal at each half-turn; next cycle seeded by polar field at the cut	compatible with the polar precursor and one-cycle memory; nothing derived
g	de Sitter (again)	A1	SHARP	see class c	thermal static patch inadmissible
g	Free diffusion / cosmic expansion in a Gaussian family	A3 no antipode	DESCR	$C = \sqrt{2} \ln \sigma$, unbounded	never cuts; irreversibility as absence of an antipode
h	Hopf point	criticality clause	CONSIST	$\rho \rightarrow 0$ with amplitude	normal form
h	Second-order thermal transition (Curie–Weiss, Ising)	criticality clause via thermal Fisher metric	TENSION	$g_{TT} = C_V/T^2$ diverges $\rightarrow \rho \rightarrow$ infinity	opposite sign to the Hopf case; the carrier-independent statement must be D8 (depth), not ρ
h	Saddle-node / critical slowing down	D8	SHARP	relaxation rate $\rightarrow 0 \Rightarrow$ Fisher speed $\sim \sqrt{k} \rightarrow 0 \Rightarrow$ depth $\rightarrow$ infinity	long memory at the transition
h	Ruppeiner geometry near criticality	A1 / D1	DESCR	$ R \sim \xi^d$	resolvable steps count correlation volumes

Reproduce: python3 crr_retrodictions.py (numpy, scipy, sympy). The script writes battery.json; this PDF is generated from it and contains no hand-typed number.